

Creating a Soil Moisture Meter

Category: Environmental Monitoring | Status: Completed Prototype | 6 min read

An Arduino-based soil moisture project that uses a capacitive sensor to understand watering needs and improve plant care decisions.

Materials

- Arduino UNO
- Capacitive soil moisture sensor
- Breadboard
- Jumper wires
- LCD or OLED display
- USB cable or battery power source

What I learned

- Calibration is what turns a sensor reading into useful information.
- Environmental data is messy because real conditions change.
- A small device can teach a lot about sensing, coding, and plant care.

Safety or accuracy note

Low-cost environmental sensors need calibration and should be treated as approximate indicators, not certified measurements.